

Female Directors and Firm Performance: Evidence under Selection on Observables

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Abstract

This paper investigates whether firms with at least one female board member perform better. The treatment is defined as having at least one female director on the board in 2015. Identification relies on the conditional independence assumption, under which treatment assignment is as good as random conditional on observables. We employ several empirical strategies, including propensity score blocking with regression adjustment, inverse probability weighting, and Double/Debiased Machine Learning (DDML), to estimate the effect on return on assets. We find positive but mostly statistically insignificant effects, with larger estimates in later years, suggesting a delayed impact. Differences across estimators indicate potential upward bias, while DDML results point to limited treatment effect heterogeneity. Overall, the evidence is consistent with a modest positive relationship, but a credible causal interpretation is hindered by potential selection on unobservables, as firms with female board representation may systematically differ in ways that also affect performance.

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1 DATA

The dataset consists of UK firms observed around the 2015 policy change introducing a soft recommendation to increase female board representation. Treatment is defined as an indicator equal to one if a firm had at least one female board member in 2015. In our sample, 131 firms are treated and 203 serve as controls. To avoid anticipation effects and ensure that covariates are not affected by treatment, we use pre-treatment characteristics from 2010 to control for selection into treatment. Outcomes are returns on assets (ROA) over 2015–2019. Descriptive statistics are reported in [Table 1](#).

TABLE 1: Summary Statistics of Main Variables

	Mean	Median	SD	Min	Max	N obs
Panel A: Treatment						
Female on Board (2015)	0.392	0.000	0.489	0	1	334
Panel B: Outcome Variables						
Return on Assets (2015)	-0.322	4.296	22.202	-145.757	44.316	334
Return on Assets (2016)	-1.430	3.867	23.664	-155.880	44.580	334
Return on Assets (2017)	-0.643	5.051	43.629	-533.777	51.392	334
Return on Assets (2018)	0.562	4.223	21.340	-162.066	58.003	334
Return on Assets (2019)	-1.943	3.310	27.814	-290.197	100.707	334
Panel C: Selected Control Variables						
Revenue (millions)	161.023	33.301	617.144	0.017	10103.625	334
Total Assets (millions)	248.141	75.079	727.448	0.372	11125.688	334
Employees	815.680	115.000	3596.920	0.000	58623.000	334
Board Size	5.784	5.000	1.639	2.000	14.000	334
ESG Score	34.736	35.305	4.223	4.976	70.494	334

Note: This table reports summary statistics for the main variables. Panel A shows the treatment indicator (female on board in 2015). Panel B reports ROA by year. Panel C includes selected firm-level controls.

To construct the analysis sample, we drop observations with missing treatment status. Missing covariates are median-imputed. Continuous variables are winsorized at the 1st and 99th percentiles to limit outliers potentially driven by misreporting or data errors. Variables are rescaled and log-transformed where appropriate.

Given the rich set of pre-treatment covariates, we apply Double Selection LASSO to select predictors of either the outcome or treatment, reducing specification bias. Using these variables, complemented by economically relevant controls, we estimate firm-level propensity scores¹. To ensure common support, we trim observations outside the [0.1, 0.9] range and re-estimate the scores². The final sample consists of 334 firms ([Table 2](#)).

¹The full list of covariates is reported in [Table A1](#).

²We rely on propensity score methods rather than direct matching, as they typically exhibit better finite-

TABLE 2: Distribution of the Estimated Propensity Score Before and After Trimming

Sample	Group	Observations	Mean	SD	Min	Max
Before trimming	Full sample	437	0.417	0.290	0.000	1.000
Before trimming	Treated Firms	183	0.619	0.285	0.074	1.000
Before trimming	Control Firms	254	0.272	0.189	0.000	0.950
After trimming	Full sample	334	0.391	0.238	0.016	0.999
After trimming	Treated Firms	131	0.537	0.235	0.064	0.999
After trimming	Control Firms	203	0.297	0.188	0.016	0.896

Note: The table reports summary statistics for the estimated propensity score before and after trimming. The top panel is based on the full sample, while the bottom panel reports the distribution after excluding observations with propensity scores outside the $[0.1, 0.9]$ range and re-estimating the scores. Treated firms are those with female board representation in 2015; control firms are those without. The distribution of propensity scores is additionally illustrated in Appendix [Figure A1](#).

2 EMPIRICAL STRATEGY

We estimate the effect of female board representation on firm performance using propensity score methods. Our main approach is stratification (blocking) on the estimated propensity score. Within each block m , we estimate:

$$Y_{it} = \alpha_m + \tau_m W_i + X_i' \beta + \varepsilon_{it}, \quad i \in m, \quad (1)$$

and aggregate block-specific estimates as

$$\hat{\tau}_{ATT} = \sum_{m=1}^M \frac{N_{m,treated}}{N_{treated}} \hat{\tau}_m. \quad (2)$$

Here, Y_{it} is ROA over 2015–2019, W_i indicates female board representation in 2015, and X_i includes pre-treatment covariates (measured in 2010). Firms are partitioned into M blocks (six in our case). The coefficient τ_m captures within-block effects, and the overall ATT is a weighted average across blocks.

We report both pure blocking and regression-adjusted (mixed) specifications, focusing on the latter due to its robustness to misspecification ([Imbens and Wooldridge, 2009](#); [Imbens, 2015](#)). As robustness checks, we also implement inverse probability weighting and the doubly robust estimator³, but do not rely on them for our main conclusions.

As an additional approach, we apply Double/Debiased Machine Learning (DDML) in a partially linear framework ([Chernozhukov et al., 2018](#); [Athey and Imbens, 2019](#); [Mul-](#)

sample performance and are straightforward to implement ([Dehejia and Wahba, 1999](#); [Busso et al., 2014](#)).

³We implement inverse probability weighting using weights $W_i / \hat{p}(X_i) + (1 - W_i) / (1 - \hat{p}(X_i))$. The doubly robust estimator augments this with an outcome regression, combining weighting and regression adjustment ([Robins and Rotnitzky, 1995](#)).

lainathan and Spiess, 2017). Unlike matching-based estimates (ATT), DDML identifies the average treatment effect (ATE) under the conditional independence assumption. We use pre-treatment covariates to flexibly control for confounding and implement 5-fold cross-fitting with stratification by treatment status. The effect is estimated via residual-on-residual regression, using multiple learners (tree-based models, neural networks, and regularized linear models) for the treatment and outcome models.

For the nuisance components, we consider several machine learning methods, including Lasso (logistic and linear), CatBoost (classifier and regressor), and a feed-forward neural network. The neural network incorporates categorical features via embedding layers and combines them with pre-treatment covariates in a simple fully connected architecture.⁴ This set of learners allows for flexible, data-driven adjustment and reduces sensitivity to model specification.

Our identification relies on the conditional independence assumption (CIA). While not testable, we assess its plausibility via covariate balance. Appendix Table A1 shows large pre-treatment differences that mostly disappear after adjustment⁵. This suggests improved comparability across groups. However, differences in unobservables may persist. In particular, firms with female board representation may systematically differ in managerial quality, corporate culture, or openness to innovation, which may also affect performance. Therefore, even under CIA-based methods, our estimates may still be subject to residual bias and should be interpreted with caution.

3 RESULTS

The main results from propensity score blocking and the regression-adjusted (mixed) specification are reported in Table 3. Estimated effects are positive throughout 2015–2019 but generally not statistically significant. Coefficients increase over time in both specifications, suggesting a delayed effect of treatment.

In 2015, the estimated effect is positive but not statistically significant (1.830), corresponding to approximately 40% of the median ROA in 2015. By 2019, the estimated effect increases to 4.450 and becomes statistically significant at the 10% level. This magnitude is economically meaningful relative to the distribution of ROA in 2019 (mean: -1.943; median: 3.310; see Table 1), amounting to approximately 130% of the median.

⁴The neural network consists of a simple fully connected architecture with two hidden layers, consisting of 6 and 3 neurons, respectively.

⁵The only remaining significant difference is log market capitalization, likely because firm size is strongly related to both board composition and firm performance, and may not be fully balanced by the matching procedure.

TABLE 3: Blocking and Blocking with Regression Adjustment Estimates of the Effect of Female Board Representation on Firm Performance

	Blocking					Blocking and Regression				
	ROA 2015	ROA 2016	ROA 2017	ROA 2018	ROA 2019	ROA 2015	ROA 2016	ROA 2017	ROA 2018	ROA 2019
Female on Board	2.139 (2.640)	3.511 (2.900)	4.591 (3.232)	4.594* (2.614)	4.234 (2.950)	1.830 (2.164)	2.952 (2.088)	4.720 (3.277)	4.451* (2.482)	4.450* (2.478)
Controls	No	No	No	No	No	Yes	Yes	Yes	Yes	Yes
Blocking	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	334	334	334	334	334	334	334	334	334	334

Note: The table reports ATT estimates of female board representation in 2015 on ROA (2015–2019) using propensity score blocking. Columns 1–5 show blocking estimates; columns 6–10 additionally include regression adjustment with pre-treatment covariates. Standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Across specifications, simple blocking yields slightly larger coefficients on average, while the regression-adjusted version produces smaller standard errors. This pattern suggests that including additional covariates improves precision and captures residual variation not accounted for by blocking alone.

To further assess the robustness of these findings, we turn to alternative estimators. Weighting-based estimates (Table 4) are generally larger in most cases than the blocking results. For example, the estimated effect on ROA in 2019 reaches 7.843 and is statistically significant at the 1% level. After including controls, the weighting estimates become closer to those obtained in Table 3.

TABLE 4: Inverse Probability Weighted Regression Adjustment Estimates of the Effect of Female Board Representation on Firm Performance

	Weighting and Regression					Weighting and Regression with Controls				
	ROA 2015	ROA 2016	ROA 2017	ROA 2018	ROA 2019	ROA 2015	ROA 2016	ROA 2017	ROA 2018	ROA 2019
Female on Board	5.451** (2.538)	5.996** (2.646)	11.095*** (4.208)	7.112*** (2.160)	7.843*** (2.861)	2.179 (2.719)	2.791 (2.324)	4.535 (3.110)	4.682 (2.871)	3.791 (2.850)
Controls	No	No	No	No	No	Yes	Yes	Yes	Yes	Yes
Weighting	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	334	334	334	334	334	334	334	334	334	334

Note: The table reports inverse probability weighted regression adjustment estimates of the ATT of female board representation in 2015 on ROA (2015–2019). Columns 1–5 report specifications without controls, while columns 6–10 include the full set of pre-treatment firm and board characteristics. Standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

This pattern points to potential selection bias. Since CIA-based methods rely on selection on observables, they cannot fully address bias from unobserved characteristics. The larger effects obtained under less restrictive specifications suggest that upward bias may be present. Importantly, this concern cannot be ruled out even for our preferred specification, as bias from unobservables is inherently untestable.

At the same time, part of the difference may reflect the finite-sample properties of weighting estimators. Inverse probability weighting is sensitive to extreme propensity score weights, and even after trimming, residual model misspecification or outliers can

inflate estimates. This further motivates our focus on blocking and regression-adjusted specifications, which are less sensitive to such issues. Overall, the results should be interpreted as suggestive rather than definitive evidence of causal effects.

We complement our analysis with Double/Debiased Machine Learning (DDML); results are reported in [Table 5](#). This approach also allows us to compare estimands across methods. While naive differences and DDML estimate the average treatment effect (ATE), blocking and weighting methods target the ATT. Despite this distinction, the estimated effects are broadly similar, suggesting limited evidence of treatment effect heterogeneity.

TABLE 5: DDML Estimates of the Effect of Female Board Representation on Firm Performance

	ROA 2015			ROA 2019		
	CatBoost	Neural Network	Lasso	CatBoost	Neural Network	Lasso
Female on Board	3.237 (1.992)	3.387* (2.039)	2.167 (1.871)	4.934* (2.975)	2.627 (3.192)	4.393* (2.263)
P-value	0.104	0.097	0.247	0.097	0.411	0.052
R ²	0.008	0.009	0.004	0.007	0.002	0.006

Note: This table reports double/debiased machine learning (DDML) estimates of the effect of female board representation on firm performance ([Chernozhukov et al., 2018](#)). Columns 1 through 3 present results for return on assets in 2015, while columns 4 through 6 report results for 2019. Treatment effects are estimated using alternative machine learning methods for nuisance parameter estimation, including CatBoost, a neural network, and Lasso. Predictive performance of these models is reported in Appendix [Table A2](#). Standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

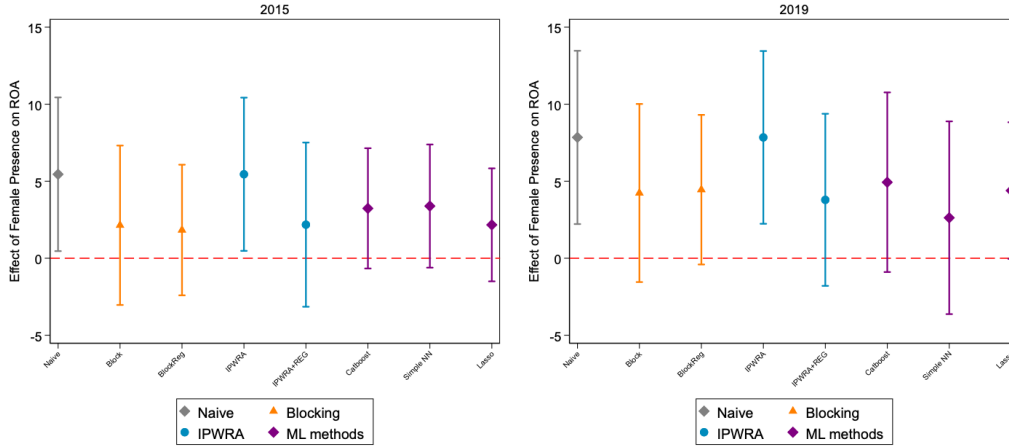
DDML estimates vary across learners. In 2015, Lasso yields more conservative estimates, while CatBoost and the neural network produce similar positive and borderline significant effects. For ROA in 2019, the neural network delivers more conservative estimates, whereas CatBoost and Lasso produce economically meaningful and statistically significant effects: 4.934 (p-value 0.097) and 4.393 (p-value 0.006), respectively.

The summary of estimated effects across specifications is presented in [Figure 1](#) (following [Gordon et al. \(2017\)](#)).

4 ROBUSTNESS CHECKS AND DISCUSSION

We conduct several robustness checks to assess the validity of our identification strategy. First, under CIA-based methods, treatment should not affect pre-treatment outcomes. We therefore estimate our main specification on pre-treatment ROA (2013 and 2014). As shown in [Table 6](#), the estimated effects are smaller in magnitude than the main estimates, statistically insignificant, and less precisely estimated. This supports the plausibility of the unconfoundedness assumption, although it cannot be directly tested.

Figure 1: Estimated Effects of Female Board Representation on Firm Performance (2015 & 2019)



Note: The figure summarizes estimated treatment effects of female board representation in 2015 on ROA. The left panel shows effects in 2015, and the right panel shows effects in 2019. Points represent estimates with 95% robust confidence intervals; the red dashed line indicates zero. Methods include: (1) Naive (difference in means), (2) Blocking (with and without regression adjustment), (3) IPWRA (without and with controls), and (4) ML-based DDML estimates (CatBoost, neural network, Lasso).

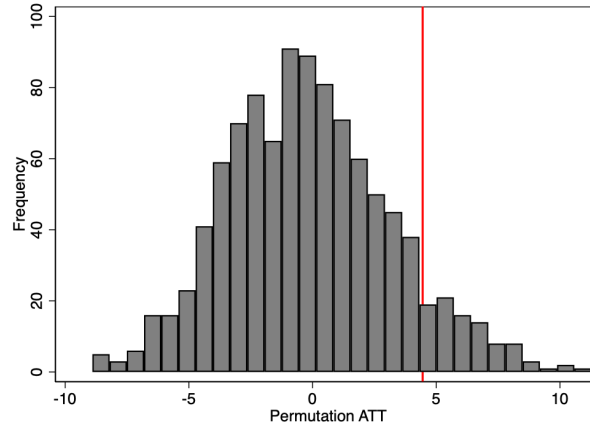
Second, we perform permutation inference for one of our main results: a 4.450 increase in ROA in 2019 (Table 3). We randomly reassign treatment within propensity score blocks, preserving the number of treated firms in each block, and re-estimate the model. As shown in Figure 2, the observed estimate lies in the tail of the permutation distribution, suggesting that the result is unlikely to be driven by chance.

TABLE 6: Placebo Tests: Female Board Representation and Pre-Treatment Firm Performance

	Blocking		Blocking and Regression Adjustment	
	ROA 2013	ROA 2014	ROA 2013	ROA 2014
Female on Board	1.202 (3.198)	-0.243 (3.490)	0.864 (2.452)	-0.619 (3.034)
Controls	No	No	Yes	Yes
Blocking	Yes	Yes	Yes	Yes
Observations	334	334	334	334
R-squared	0.015	0.054	0.155	0.404

Note: The table reports placebo estimates of the effect of female board representation in 2015 on pre-treatment ROA (2013–2014). Columns 1–2 present blocking estimates, while columns 3–4 include regression adjustment with pre-treatment covariates. As outcomes precede treatment, estimated effects capture pre-treatment differences rather than causal impacts. Standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 2: Permutation Inference for the Effect of Female Board Representation



Note: The figure shows the permutation distribution of treatment effects on ROA in 2019. Treatment is randomly reassigned within propensity score blocks, preserving the number of treated firms in each block, and the blocking with regression adjustment specification is re-estimated in each replication. The vertical line indicates the observed estimate.

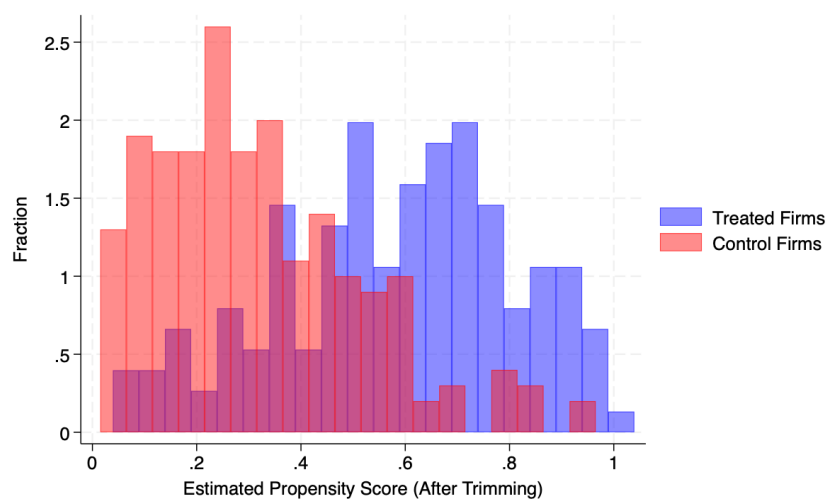
5 CONCLUSION

This paper examines whether female board representation improves firm performance among UK-listed companies following the 2015 soft policy recommendation. Using propensity score blocking with regression adjustment as our preferred estimator, we find a positive but largely statistically insignificant effect on return on assets. The effect becomes marginally significant only in 2018–2019, suggesting a delayed impact. Estimates are broadly consistent across methods (blocking, IPWRA, and DDML), indicating limited treatment effect heterogeneity. Permutation inference further suggests that the 2019 result is unlikely driven by chance.

These findings should be interpreted with caution. The conditional independence assumption (CIA) is untestable, and unobserved factors such as managerial quality, corporate culture, and governance practices may jointly affect board composition and performance. The larger effects obtained from less restrictive estimators point to potential upward bias. While placebo tests provide some support for the identification strategy, they cannot fully rule out this concern. Overall, the evidence is consistent with a modest positive effect, but a causal interpretation requires stronger exogenous variation in board composition.

6 APPENDIX

Figure A1: P-score Overlap Histogram



Note: The figure shows the distribution of the re-estimated propensity score after trimming observations outside the selected support region. The histogram is reported separately for treated and control firms. Overlap between the distributions indicates the region of common support used in subsequent estimation.

TABLE A1: Balance of Pre-treatment Characteristics Before and After Blocking

	Raw Means			Controlling for Blocks		
	Treated Firms	Control Firms	Difference	Treated Firms	Control Firms	Difference
Return on Assets	8.292 (1.660)	1.739 (1.610)	6.553*** (2.312)	4.478 (1.731)	4.201 (1.608)	0.277 (2.474)
Log Total Revenue	18.045 (0.163)	16.794 (0.130)	1.251*** (0.209)	17.312 (0.155)	17.267 (0.110)	0.045 (0.201)
Log Market Capitalization	18.895 (0.144)	17.473 (0.103)	1.421*** (0.177)	18.223 (0.122)	17.907 (0.083)	0.316** (0.154)
ESG Score	34.548 (0.492)	34.857 (0.209)	-0.310 (0.535)	34.801 (0.385)	34.694 (0.276)	0.107 (0.468)
Independent Directors Share	0.526 (0.026)	0.419 (0.019)	0.106*** (0.032)	0.459 (0.025)	0.463 (0.019)	-0.004 (0.033)
Director Network Size	850.896 (54.407)	829.381 (47.154)	21.516 (71.998)	838.607 (56.822)	837.312 (48.358)	1.295 (76.984)
Board Age	56.700 (0.417)	56.528 (0.367)	0.172 (0.555)	56.568 (0.499)	56.613 (0.372)	-0.045 (0.667)
Board Age Squared	3237.441 (46.865)	3222.544 (40.860)	14.897 (62.176)	3225.824 (55.666)	3230.041 (41.351)	-4.217 (74.065)
Number of Director Qualifications	1.528 (0.051)	1.490 (0.039)	0.038 (0.064)	1.495 (0.057)	1.512 (0.040)	-0.017 (0.074)
Board Size	6.221 (0.147)	5.502 (0.109)	0.719*** (0.183)	5.805 (0.145)	5.771 (0.111)	0.033 (0.195)
Board Size Squared	41.534 (2.143)	32.655 (1.303)	8.879*** (2.509)	36.275 (1.959)	36.049 (1.407)	0.227 (2.549)
Net Income	24.103 (4.518)	6.729 (1.515)	17.374*** (4.765)	13.931 (3.215)	13.293 (1.988)	0.638 (3.640)
Return on Equity	8.146 (3.530)	3.708 (2.813)	4.439 (4.514)	4.985 (3.246)	5.748 (3.014)	-0.763 (4.522)
Number of Employees	1459.298 (487.358)	400.340 (63.923)	1058.958** (491.532)	957.802 (296.490)	723.965 (149.381)	233.837 (233.302)
Sector: Basic Materials	0.084 (0.024)	0.089 (0.020)	-0.005 (0.031)	0.089 (0.027)	0.085 (0.021)	0.004 (0.037)
Sector: Consumer Cyclicals	0.107 (0.027)	0.099 (0.021)	0.008 (0.034)	0.103 (0.031)	0.101 (0.023)	0.002 (0.042)
Sector: Consumer Non-Cyclicals	0.053 (0.020)	0.039 (0.014)	0.014 (0.024)	0.050 (0.017)	0.042 (0.013)	0.009 (0.020)
Sector: Energy	0.053 (0.020)	0.069 (0.018)	-0.016 (0.027)	0.063 (0.021)	0.063 (0.017)	0.000 (0.027)
Sector: Financials	0.351 (0.042)	0.281 (0.032)	0.070 (0.052)	0.309 (0.043)	0.308 (0.034)	0.000 (0.057)
Sector: Healthcare	0.053 (0.020)	0.084 (0.020)	-0.030 (0.028)	0.061 (0.026)	0.079 (0.021)	-0.017 (0.038)
Sector: Industrials	0.160 (0.032)	0.187 (0.027)	-0.027 (0.042)	0.174 (0.035)	0.178 (0.028)	-0.004 (0.048)
Sector: Technology	0.130 (0.029)	0.143 (0.025)	-0.013 (0.038)	0.143 (0.034)	0.134 (0.026)	0.008 (0.045)
Operating Income	30.577 (4.416)	8.587 (1.638)	21.990*** (4.710)	17.879 (2.975)	16.782 (1.932)	1.097 (3.461)
Total Dividends Paid	4.756 (0.587)	1.797 (0.281)	2.959*** (0.651)	2.993 (0.437)	2.935 (0.336)	0.058 (0.585)
Long Term Debt	46.159 (9.829)	14.033 (2.964)	32.126*** (10.266)	28.076 (6.224)	25.702 (4.468)	2.374 (7.246)

Note: The table reports balance in pre-treatment firm and board characteristics between treated and control firms. Columns 1–3 show raw means and differences. Columns 4–6 report means and differences after controlling for propensity score blocks. Block-adjusted differences are obtained from regressions of each covariate on the treatment indicator and block fixed effects; reported differences correspond to the coefficient on the treatment indicator. Standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A2: Performance of Machine Learning Models

	Treatment Probability			ROA 2015		ROA 2019	
	Accuracy	F1 score	ROC AUC	RMSE	MAE	RMSE	MAE
CatBoost	0.7155	0.6452	0.7731	15.4871	9.2519	25.2839	11.4171
Neural Network	0.7069	0.6623	0.7435	16.3842	10.0050	24.6234	12.1544
Lasso	0.6897	0.6250	0.7185	16.6678	9.9229	25.5723	11.7165

Note: The table reports predictive performance of machine learning models used in the DDML procedure. Columns 1-3 present classification metrics for the treatment model, where higher values indicate better performance. Columns 4-5 and 6-7 report prediction errors for outcome models in 2015 and 2019, where lower values indicate better accuracy. All metrics are based on out-of-fold predictions from a 5-fold cross-fitting procedure with stratification by treatment status.

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